MGU – UGP (HONOURS) REGULAR Examination OCTOBER 2025

THIRD SEMESTER

QP CODE: 25900357 – DSC – DIGITAL FUNDAMENTALS – MG3DSCECT204

ANSWER SCHEME

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

PART A

Multiple choice questions

Answer all questions. Each question carries 1 marks

1) c – 162
2) d – D
3) b – To handle interrupt requests and prioritize them
4) a – 1
5) c – Product of the compliments
6) c – flexibility in instruction design
7) a – A+BC
8) a – Immediate
9) b – Peripherals access memory directly without CPU
10) a – D7

PART B

Short Answer Questions

Answer any 4 questions. Each question carries 5 marks

11. Explanation of 2's compliment with an example any method
    Explanation - 2.5 marks
    Example - 2.5 marks
12. Explanation of volatile memory and non-volatile memory
    Volatile memory explanation and different types - 2.5 marks
    Non-volatile memory explanation different types – 2.5 marks
13. Explanation of MAR and MDR - 2.5 marks each
14. Half adder explanation – 1 mark

Truth table and Equation – 2 marks

Circuit – 2 marks

15. Explanation of PIPO - 2 marks

    Diagram - 3 marks

16. Any 5 points - each carries 1 mark

PART C

Essay Questions

Answer any 2 questions. Each question carries 10 marks

17. Explanation of BCD numbers with example – 5 marks

    Advantages – 2 marks

    Disadvantages – 2 marks

    Applications – 1 mark

18. Decoder explanation – 4 marks

    Truth table – 3 marks

    Circuit – marks

19. Explanation of cache memory mapping - 5 marks

    Direct mapping – 2 marks

    Associative mapping – 2 marks

    Set associative mapping – 2 marks